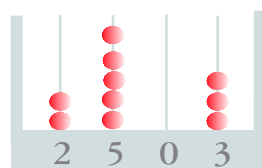




INDIAN *numbers*

In ancient times, the best way of doing sums was with an **abacus** – a calculating device made of rows of beads or stones. But about 1500 years ago, people in India had a better idea. They invented a “place



system” – a way of writing numbers so that the symbols matched the rows on an abacus. This meant you could do tricky sums **without** an abacus, just by writing numbers down. A symbol was needed for an empty row, so the Indians invented **zero**. It was a stroke of genius.

The new numbers spread from Asia to Europe and became *the numbers we use today*.

Unlike other number systems, the Indian system had only **10 symbols**, which made it wonderfully simple. These symbols changed over the centuries as they spread from place to place, gradually evolving into the modern digits we all now use.

300 BC to 400 AD	400 AD to 600 AD	700 AD to 1100	900 AD to 1200	16th century
—	—	१	1	1
=	=	२	2	2
≡	≡	३	3	3
𑂔	𑂔	४	4	4
𑂕	𑂕	५	5	5
𑂖	𑂖	६	6	6
𑂗	𑂗	७	7	7
𑂘	𑂘	८	8	8
𑂙	𑂙	९	9	9
		०	0	0

EUROPE 1200 to NOW Indian numbers slowly replaced Roman numbers in Europe as people discovered how useful they were for calculating. The new numbers helped trigger the Renaissance, or “age of learning” – the period of history in which modern science was born.

ENGLAND 1100 AD

Adelard of Bath, an English monk, visited North Africa disguised as an Arab. He translated Al Khwarizmi's books and brought zero back to England. As he only told other monks, nothing happened.

NORTH AFRICA 1200 AD

Indian numbers were picked up by Italian merchants visiting the Arab countries of North Africa. In 1202 an Italian called Fibonacci explained how the numbers worked in a book called *Liber Abaci*, and so helped the Indian system spread to Italy.